

Parag Wadhai

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AI/ML Engineer with 1+ year of experience designing, developing, and deploying production-grade AI, Machine Learning, and Generative AI solutions. Specialized in Large Language Models (LLMs), Prompt Engineering, RAG, Real-Time Analytics, and Multimodal AI, with hands-on experience building intelligent systems for telecom operations, automated Root Cause Analysis (RCA), anomaly detection, and image analysis. Proficient in Python, FastAPI, Kafka, ClickHouse, and modern AI frameworks, with a proven ability to develop scalable, data-driven applications that enhance operational efficiency, automate complex workflows, and deliver actionable business insights.

Skillsets

Programming & Tools: Python, SQL, FastAPI, Git, Docker, Linux, Postman, REST APIs, Async Programming, Multithreading

AI/ML & GenAI: Machine Learning, Deep Learning, LLMs, Prompt Engineering, NLP, CNN, Scikit-learn, TensorFlow, Feature Engineering, Model Evaluation, Hyperparameter Tuning, LangChain, LangGraph, RAG

LLMs & AI Models: Qwen2.5-VL, Gemma 3/4, Llama, Claude

Data Analytics: Data Analysis, Statistical Analysis, KPI Analytics, Network Analysis, Exploratory Data Analysis (EDA), Predictive Analytics

Data Engineering: Apache Kafka, ClickHouse, Real-Time Data Pipelines, Stream Processing, Event-Driven Architecture, Data Ingestion Pipelines

Databases: PostgreSQL, MySQL, MongoDB, ClickHouse

Professional Experience

Amantya Technologies

AI/ML Engineer | Nagpur, India

Jul 2025 – Present

- AI-Powered Network Operations Analysis:** Designed and developed an intelligent telecom operations platform using **Qwen2.5-VL-7B-Instruct**, prompt engineering, and NLP techniques to automate **Root Cause Analysis (RCA)**, log interpretation, and report generation, reducing manual troubleshooting effort by **70%**.
- Large-Scale Log Intelligence:** Processed and analyzed **10K+ telecom network logs daily**, enabling faster fault identification and improving operational visibility across multiple network components.
- Telecom Infrastructure Image Analysis:** Built a multimodal image inspection workflow using **Gemma-4-31B-IT**, automating telecom tower assessment and defect detection from field images, reducing inspection turnaround time by **60%**.
- Real-Time Network Analytics Platform:** Developed scalable **FastAPI microservices** and **Kafka-based streaming pipelines** capable of processing thousands of events per minute for real-time network monitoring and analytics.
- Concurrent AI Processing:** Implemented **multithreading and asynchronous processing** to execute image analysis, log analysis, database operations, and API workflows concurrently, reducing end-to-end processing latency by **40%**.
- Operational Data Intelligence:** Leveraged ClickHouse to store and analyze **large-scale telecom operational datasets**, improving RCA retrieval performance by **50%**.
- Production AI Deployment:** Integrated **LLM-powered** workflows into production environments, supporting automated anomaly investigation, network diagnostics, and predictive analysis.

Amantya Technologies

Engineer Trainee | Nagpur, India

Jun 2025 – Jan 2025

- Network Anomaly Detection System:** Developed anomaly detection models using **One-Class SVM, SVM**, and statistical techniques to identify abnormal network behavior, achieving **85%+ anomaly detection accuracy**.
- PCAP KPI Analytics Engine:** Built a high-performance analytics framework to process **10K+ network packets per session**, extracting KPIs including throughput, latency, jitter, variance, and packet loss.
- Feature Engineering & Optimization:** Improved model performance by **15–20%** through advanced feature engineering, data preprocessing, and hyperparameter tuning using **GridSearchCV**.
- Backend Analytics Integration:** Integrated machine learning models into backend services to enable automated anomaly detection and real-time network performance monitoring.
- Parallel Data Processing:** Utilized **multithreading and concurrent processing techniques** for KPI extraction and analytics workloads, reducing processing time by **35%**.
- Data Pipeline Development:** Built automated data ingestion and processing workflows for large-scale telecom datasets, improving analysis efficiency and reducing manual intervention.

Technocolabs Software

Machine Learning Engineer Trainee | Remote

Oct 2024 – Dec 2024

- Employee Attrition Analytics:** Developed machine learning models to analyze workforce attrition trends using classification and predictive analytics techniques.
- Reporting Automation:** Automated **20+ business reporting workflows**, reducing report generation time by **30%** and improving reporting consistency.
- Predictive Modeling:** Designed and implemented complete machine learning pipelines covering data collection, preprocessing, model training, evaluation, and deployment, improving data processing efficiency by **50%**.
- End-to-End ML Pipelines:** Designed end-to-end machine learning pipelines, improving data processing efficiency by **50%**.
- Performance Optimization:** Applied data engineering and model optimization techniques to reduce model training and inference times by **25%**.

Education

Priyadarshini College of Engineering, Nagpur

Aug 2021 - July 2024

B.Tech in Artificial Intelligence and Data Science

8.30 CGPA